

Python 01

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“The magic of computing begins with 0, a simple binary digit that serves as a powerful reminder that even the smallest building blocks can create wonders.”

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Introduction

Starting to code

This course is designed to give you a basic understanding of the Python programming language. But to get you all up to speed we will first have a look at the Tools you need to get started...

Tools for the course

- Python - python.org/downloads
- VsCode - code.visualstudio.com/download
- COMPUTOR - [Getting started Guide - Computer](#)
- COMPUTOR - [Computer extension download-link](#)

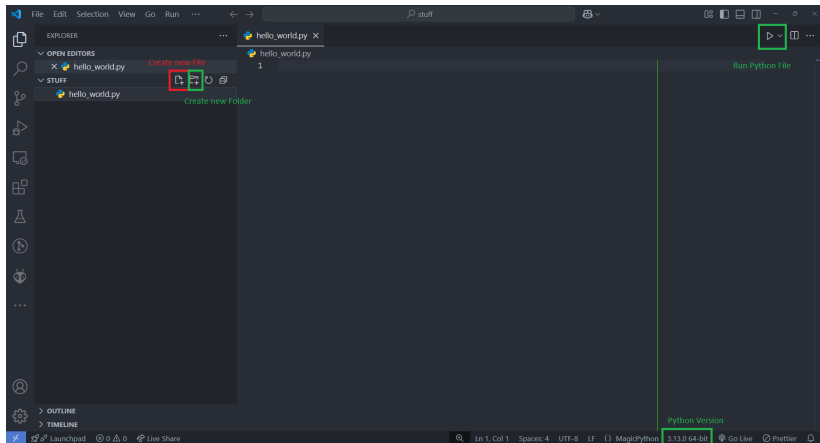
Python & VsCode

Python in VsCode

Ok, so you have installed Python and VsCode.

Let's take a look ...

Python in VsCode



Hello World!

```
1 # This is a comment in Python
2
3 # Printing to the console
4 print("Hello, world!")
```

After we have written the code we click on the run button in the top right corner.

We get the following output in the terminal:

```
C:\REPOS\exercises-python>"C:/Program Files/Python313/python.exe" c:/REPOS/exercises-python/slides25/01/examples/hello_world.py
Hello, world!
```

Variables: Math vs. Programming

Mathematics:

$x = 5 \Rightarrow x$ always stays 5

Programming:

```
1 x = 5
2 x = x + 1
3 print(x)    # Output: 6
```

$x = 5 \Rightarrow$ value of x can be changed

Defining variables

```
1 # Variables and data types
2 name = "Alice"    # String
3 age = 30          # Integer
4 height = 175.5    # Float
5 stud_b = True     # Boolean
```

And we can also print the variables to the terminal.

```
1 print("Name:", name) # Name: Alice
2 print("Age:", age)   # Age: 30
3 print("Height:", height) # Height: 175.5
4 print("Student?:", stud_b) # Student?: True
```

Python as a calculator

```
1 # Basic arithmetic operations
2 sum_ = age + 5
3 difference = height - 1.5
4 product = age * 2
5 quotient = height / 2
6
7 print("Sum:", sum_) # 35
8 print("Difference:", difference) # 4.0
9 print("Product:", product) # 60
10 print("Quotient:", quotient) # 2.75
```

Additional Resources

Additional Resources

- Codewars
- Leetcode
- Oh my git!
- Interactive python tutorials
- Learn Python 3
- CS50